

1.5 Chemical Energetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Medium

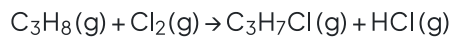
Time allowed: 60

Score: /44

Percentage: /100

Question 1a

Propane reacts with chlorine to form chloropropane.



i)

Use bond energies from Table 1.1 to calculate the enthalpy change for this reaction.

Table 1.1

Bond	Bond Energy
C-H	410
C-Cl	340
Cl-Cl	242
H-Cl	431

Include a sign in your answer.

enthalpy change = kJ mol⁻¹

[3]

ii)

State the conditions needed for this reaction to occur.

[1]

[4 marks]

Question 1b

Using Fig. 1.1, construct a labelled reaction pathway diagram for the reaction in part (a) including the activation energy.

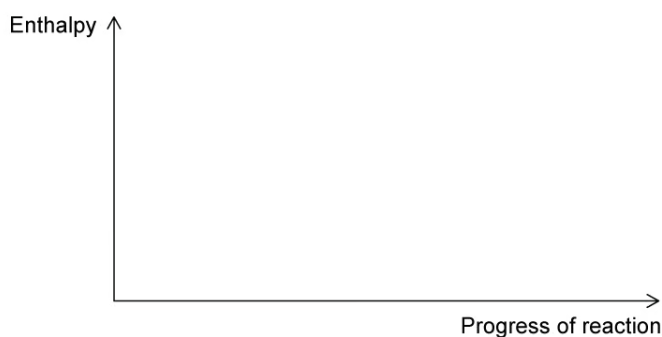


Fig. 1.1

[3 marks]

Question 1c

Ethane and chlorine will also react together under the same conditions.

State and explain the difference in enthalpy change for the two reactions

[2 marks]

Question 1d

Propane will react with bromine in a similar reaction to part (a).

Suggest, with a reason, whether the sum of the bonds broken would be larger or smaller compared to the reaction between propane and chlorine.

[2 marks]

Question 2a

The apparatus shown in Fig. 2.1 can be used to determine the enthalpy of combustion of butan-1-ol, $\text{C}_4\text{H}_9\text{OH}$ ($M_r = 74.12 \text{ g mol}^{-1}$).

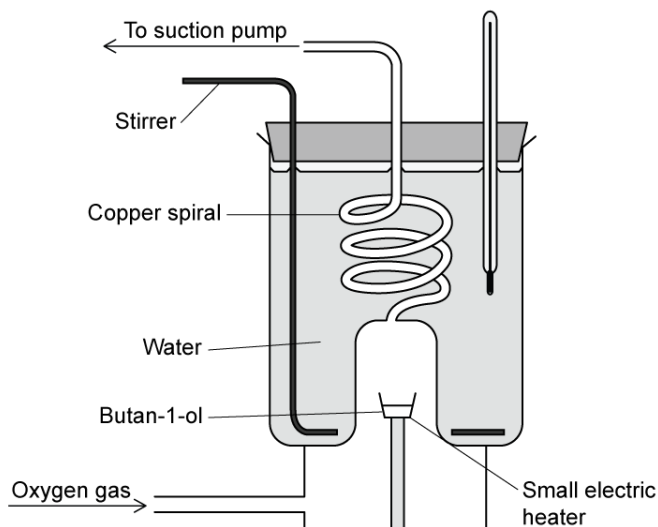


Fig. 2.1

i)

Write an equation to represent the enthalpy of combustion of butan-1-ol.

[2]

ii)

Suggest the purpose of the copper spiral and small electric heater in the apparatus.

[2]

[4 marks]

Question 2b

An experiment was carried out and the follow measurements were recorded in Table 2.1

Table 2.1

Mass of butan-1-ol / g	2.20 g	Initial temperature of water / °C	22.5
Volume of water / cm ³	875 cm ³	Final temperature of water / °C	25.0

(Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

Use the data to calculate the enthalpy of change, q , for the reaction. Give your answer to two significant figures.

$q = \dots\dots\dots \text{ kJ}$

[2 marks]

Question 2c

Determine the enthalpy of combustion of butan-1-ol using Table 2.1 and your answer to part (b).

Give your answer to three significant figures.

$\Delta H = \dots\dots\dots \text{ kJ mol}^{-1}$

[3 marks]

Question 2d

The standard enthalpy of combustion of butan-1-ol is $-332.42 \text{ kJ mol}^{-1}$.

i)

Calculate the percentage error from the experiment.

[1]

ii)

Suggest one source of error that could account for the difference between the experiment value and the standard enthalpy change.

[1]

[2 marks]

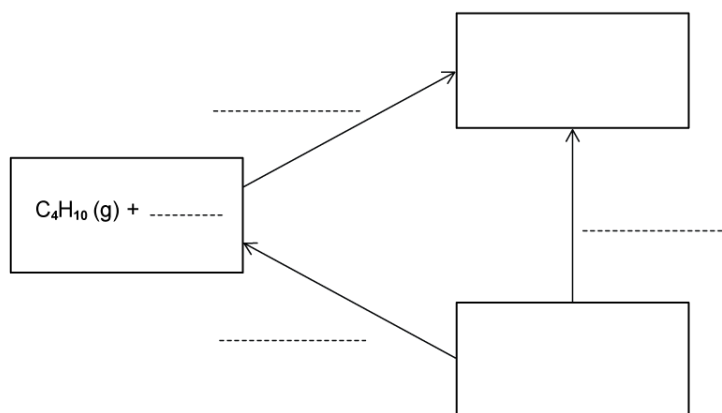
Question 3a

Complete the following Hess' Law energy cycle relating butane to enthalpies of formation and combustion.

- enthalpy of formation, ΔH_f^θ
- enthalpy of combustion, ΔH_c^θ

On your diagram:

- include the relevant species in the two empty boxes,
- label each enthalpy change with its appropriate symbol,
- complete the remaining two arrows showing the correct direction of enthalpy change



[4 marks]

Question 3bUse the data in Table 3.1 to calculate the enthalpy of combustion of butane, ΔH_c^θ **Table 3.1**

Substance	Enthalpy of formation, $\Delta H_f^\theta / \text{kJ mol}^{-1}$
$\text{CO}_2(\text{g})$	-393.5
$\text{C}_4\text{H}_{10}(\text{g})$	-125
$\text{H}_2\text{O}(\text{g})$	-242

$$\Delta H_c^\theta (\text{C}_4\text{H}_{10}) = \dots\dots\dots \text{kJ mol}^{-1}$$

[1 mark]

Question 3c

Using the values given in **Table 3.2**, calculate the enthalpy of combustion of butane.

Table 3.2

Bond	Bond Energy / kJ mol^{-1}
C-C	348
C-H	410
C=O	743
H-O	463
O=O	495

$$\Delta H_c^\theta (\text{C}_4\text{H}_{10}) = \dots\dots\dots \text{kJ mol}^{-1}$$

[1 mark]**Question 3d**

Suggest, with a reason, which of the two values of enthalpy of combustion from parts **(b)** and **(c)** is likely to be more accurate.

[2 marks]

Question 4a

Two common oxides of nitrogen are nitrogen monoxide, NO, and nitrogen dioxide, NO₂.

i)

Complete the Table 4.1 to show the oxidation number of nitrogen in each compound.

Table 4.1

Compound	NO	NO ₂
oxidation number of N		

[1]

ii)

Write equations for the formation of NO₂ by:

- The reaction of N₂ with O₂
- The reaction of NO with O₂

[2]

[1 mark]

Question 4b

Molecules of NO₂ can be formed by the reaction between N₂ and O₂. The bond between the N and O atoms in NO₂ is a double covalent bond.

The enthalpy change of reaction for this reaction is +497 kJ mol⁻¹. Calculate the bond enthalpy, in kJ mol⁻¹, of the N=O bond. Use relevant data from Table 4.2.

Table 4.2

Bond	Bond energy / kJ mol ⁻¹
N≡N	941
O=O	495

Bond enthalpy of the N=O bond = kJ mol⁻¹

[2]

[3 marks]

Question 4c

The boiling points of carbon dioxide and nitrogen dioxide are $-78.5\text{ }^{\circ}\text{C}$ and $21\text{ }^{\circ}\text{C}$ respectively.

Suggest a reason for the difference.

[2 marks]

Question 4d

Write an equation for the enthalpy of formation of nitrogen monoxide, NO.

[1 mark]

Question 5a

Propane gas is used widely as a fuel which can be used in camping gas stoves. The enthalpy of combustion for propane is $-2219.2\text{ kJ mol}^{-1}$.

What is the minimum mass of fuel is needed to bring 150 cm^3 of water at $21\text{ }^{\circ}\text{C}$ to its boiling point?

Show your working.

.....g

[3 marks]

Question 5b

Suggest why more propane may be required than calculated in part (a).

[1 mark]

Question 5c

Using Fig. 5.1, construct a labelled reaction pathway diagram for the reaction in part (a) including the activation energy.



Fig. 5.1

[3 marks]

Model Answers: Medium

1a

a)

i) The enthalpy change for this reaction is:

- Sum of the bonds broken:

$$E(\text{C-H}) = 410$$

$$E(\text{Cl-Cl}) = 242$$

$$\text{Total} = +652 \text{ kJ}; [1 \text{ mark}]$$

- Sum of the bonds formed:

$$E(\text{C-Cl}) = 340$$

$$E(\text{H-Cl}) = 431$$

$$\text{Total} = -771 \text{ kJ}; [1 \text{ mark}]$$

- Enthalpy change = $652 - 771 = -119 \text{ kJ}$; [1 mark]

ii) The conditions needed for this reaction to occur are:

- UV light / sunlight / high temperature; [1 mark]

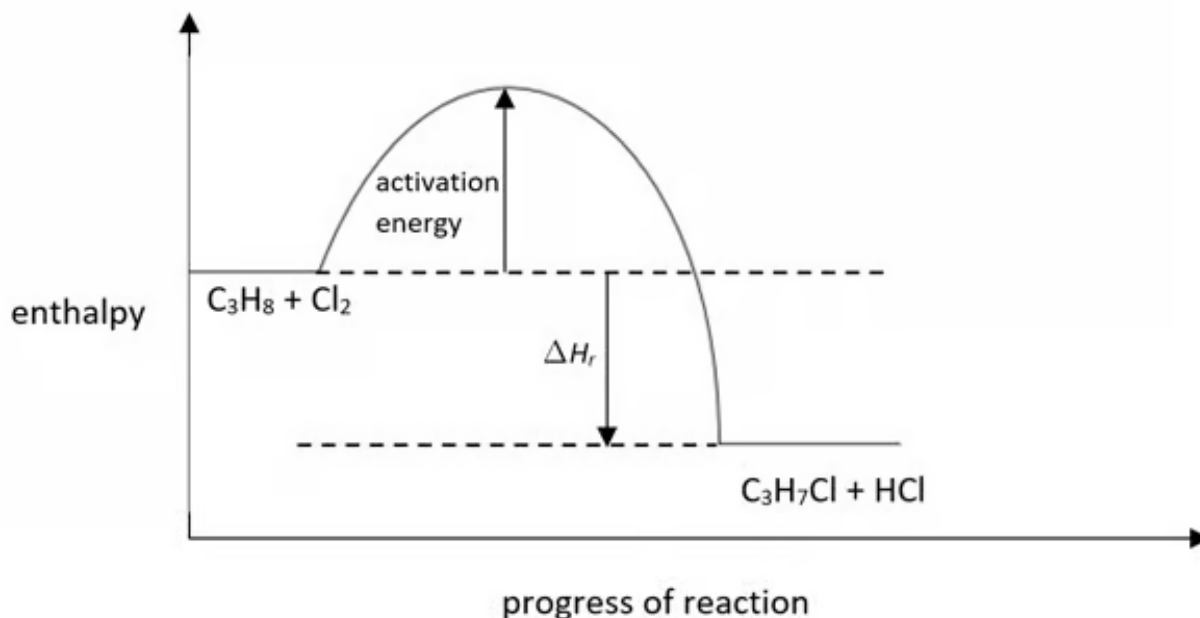
[Total: 4 marks]

- Always have a methodical approach to bond energy calculations and break it down into steps
- Be careful to check if more than one of the same bond is broken or formed
- Remember that bonds broken have a positive enthalpy change because energy is needed to break the bonds
- Find the enthalpy change of reaction using:

$$\text{Enthalpy change} = \Sigma \text{ bonds broken} + \Sigma \text{ bonds made}$$

1b

b) A labelled reaction pathway (energy level) diagram for the reaction in part (a) including the activation energy:



- Correct location of reactants and products; [1 mark]
- Upward arrow labelled as activation energy; [1 mark]
- Downward arrow labelled as ΔH_r / enthalpy change; [1 mark]

[Total: 3 marks]

- There is no need to add the values for the enthalpy changes in part a), just the relative starting and finishing positions of the reactants and products

1c

c) The difference in enthalpy change for the two reactions is:

- There is no / little difference in enthalpy change for the reactions; [1 mark]
- Because the same number and type of bonds are broken and formed in both reactions; [1 mark]

[Total: 2 marks]

- Bond energies are average values obtained from a range of similar compounds
- There may be very slight differences in bond energy between compounds as the

presence of other atoms and bonds could have a small influence on bond strength

- For this reason you could argue that there may be a small difference between the halogenation of propane and the halogenation of ethane

1d

d) Whether the sum of the bonds broken would be larger or smaller with bromine instead of chlorine:

- Smaller; [1 mark]
- Bromine is a larger atom than chlorine;

AND

The Br-Br bond is (longer and) weaker than Cl-Cl; [1 mark]

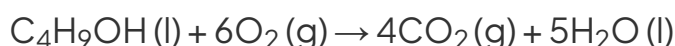
[Total: 2 marks]

- You should know the trends in atomic radius for the halogens which allows you to predict the difference in bond length between halogen molecules
- Shorter covalent bonds are stronger than longer bonds

2a

a)

i) An equation to represent the complete combustion of butan-1-ol:



- Correct products; [1 mark]
- Correct balancing of reactants and products; [1 mark]

ii) The purpose of the copper spiral and small electric heater in the apparatus:

- The copper spiral ensures complete heat exchange with the surrounding water / conducts heat from the waste gases (combustion products) to the water; [1 mark]
- To ignite (and vaporise) the butan-1-ol; [1 mark]

[Total: 4 marks]

- Be careful when you balance combustion reactions of alcohols as it is very easy to miss the oxygen in the alcohol itself

- Although you may not have seen this apparatus before you should have performed or seen an enthalpy of combustion experiment and be able to use your knowledge and apply it in unfamiliar circumstances
- You must have the idea that the alcohol is burning- only stating that it is vaporised by the heater would not be enough to score the mark as vaporising alone doesn't include combustion

2b

b) The enthalpy of change, q , for the reaction is:

- $q = mc\Delta T$;

OR

$$q = 875 \times 4.18 \times 2.5; [1 \text{ mark}]$$

- $q = 9,143.75 \text{ J} = 9,100 \text{ J} = 9.1 \text{ kJ to 2 sig figs}; [1 \text{ mark}]$

[Total: 2 marks]

- Since you are given the unit in the answer space make sure you convert to kilojoules or you may be penalised since this is a relatively simple calculation to perform
- Failing to round to two significant figures would also cost you a mark

2c

c) The enthalpy of combustion of butan-1-ol is:

- $\Delta H = \frac{q}{n}$; [1 mark]

- $n = \frac{2.20 \text{ g}}{74.12 \text{ g mol}^{-1}} = 0.02968 \text{ mol} ; [1 \text{ mark}]$

- $\Delta H = \frac{9143.75 \text{ J}}{0.02968 \text{ mol}} = 308,077.83 = -308 \text{ kJ mol}^{-1} ; [1 \text{ mark}]$

[Total: 3 marks]

- Don't round off a calculation until the end
 - If you used 9,100 J from part (b) you would get -307 kJ mol^{-1} as your final answer and would lose a mark

- A correct final answer without working would get full marks
- Failure to include the minus sign will cost you a mark

2d

d)

i) The percentage error from the experiment is:

- Percentage error = ;

$$\text{Percentage error} = \frac{332.42 - 308}{332.42} \times 100 = 7.3\%; [1 \text{ mark}]$$

ii) One source of error that could account for the difference between the experiment value and the standard enthalpy change is:

- Heat losses (to the surroundings) / incomplete combustion of the alcohol / incomplete heat transfer to the copper spiral; [1 mark]

[Total: 2 marks]

- Heat losses to the surroundings is the biggest error in combustion experiments, followed by incomplete combustion
- A long spiral slows the movement of the combustion gases enabling an efficient transfer of energy, as well as making the spiral out of a good conductor like copper
- Having a stream of oxygen passed into the apparatus help to ensure the combustion is complete
- One minor error that can be compensated for is to take into account the energy delivered by the heater
 - This is usually done by turning on the heater before the combustion, measuring the amount of joules consumed with a joulemeter (or knowing the current and voltage and time) and measuring the change in water temperature
 - This enables you to calculate the specific heat capacity of the apparatus and use it instead of 4.18

3a

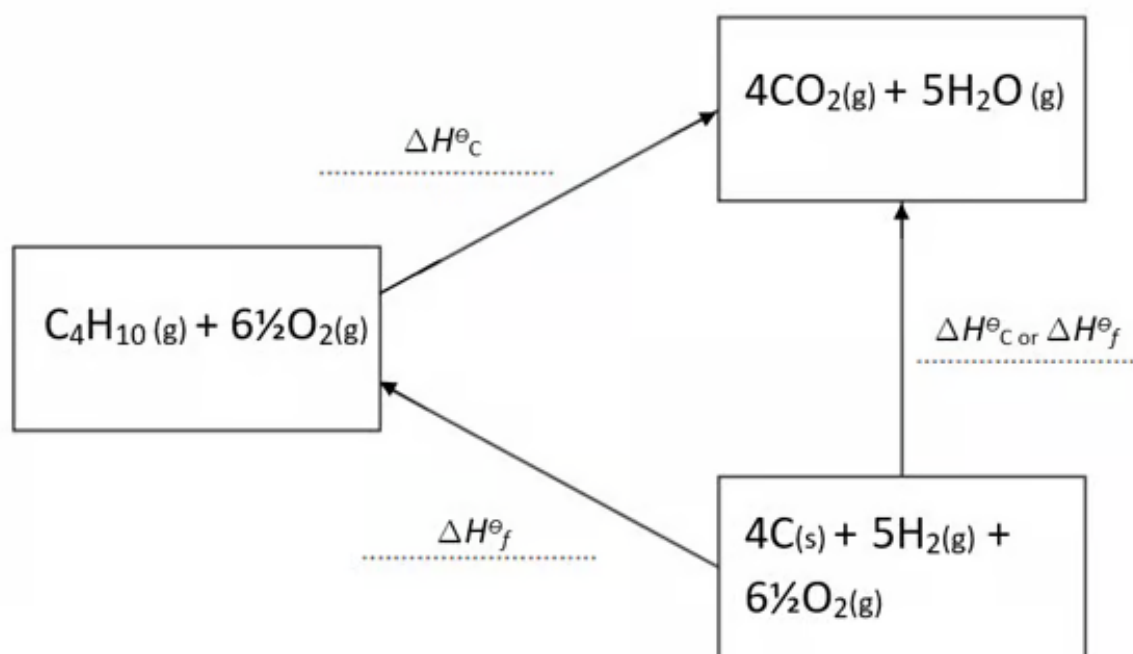
- Heat losses (to the surroundings) / incomplete combustion of the alcohol / incomplete heat transfer to the copper spiral; [1 mark]

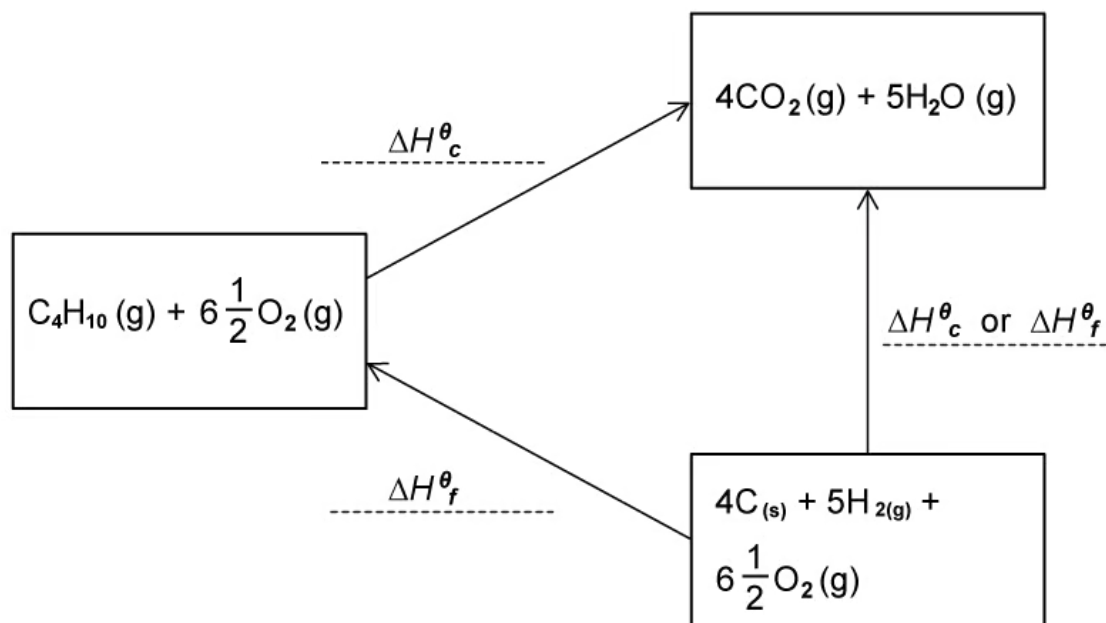
[Total: 2 marks]

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3a

a) The Hess' Law cycle for butane





- Correct formulae in the boxes; [1 mark]
- Correct balancing/ coefficient in the boxes; [1 mark]
- Correct enthalpy changes labelled; [1 mark]
- Both arrows shown in the correct direction; [1 mark]

[Total: 4 marks]

- You must include the state symbols for the substance as they are part of the standard enthalpy change definitions
- The formation of oxides is the same as the combustion of elements, so either term can be used

3b

b) Calculating the enthalpy of combustion of butane, ΔH_c^θ :

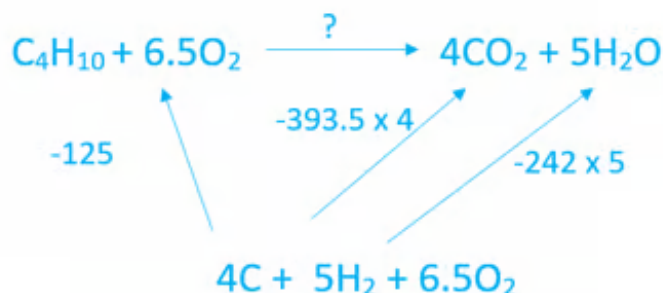
- $\Delta H_c^\theta = -\Delta H_f^\theta[\text{C}_4\text{H}_{10}(\text{g})] + 4\Delta H_f^\theta[\text{CO}_2(\text{g})] + 5\Delta H_f^\theta[\text{H}_2\text{O}(\text{g})]$; [1 mark]
- $\Delta H_c^\theta = -(-125) + 4(-393.5) + 5(-242) = -2659 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 2 marks]

- The enthalpy of combustion equation is:



- It can help to draw a Hess cycle with the values in:



- To find the unknown go around the cycle anticlockwise, add if the arrow(s) follows the direction, subtract if it goes against the direction
- Be careful with your signs- it is a good idea to separate the sign from the mathematical operation by using brackets around the terms

3c

c) The enthalpy of combustion of butane is:

Bonds Broken / kJmol^{-1}	Bonds Formed / kJmol^{-1}
$3 \times \text{C}-\text{C} = 3 \times 348 = 1044$ $10 \times \text{C}-\text{H} = 10 \times 410 = 4100$ $6.5 \times \text{O}=\text{O} = 6.5 \times 495 = 3217.5$	$8 \times \text{C}=\text{O} = 8 \times 743 = 5944$ $10 \times \text{O}-\text{H} = 10 \times 463 = 4630$
Total = +8361.5 [1 mark]	Total = -10574 [1 mark]

- $\Delta H_f = +8361.5 - 10574 = -2212.5 \text{ kJ}$; [1 mark]

[Total: 3 marks]

Sometimes it is easier to set out bond energy calculations like a balance sheet

3d

d) Of the two values of enthalpy of combustion from parts **(b)** and **(c)** which is likely to be more accurate:

- The value from part (b) / -2659 kJ ; [1 mark]
- The calculation in (b) uses data from experimentally measured values;

AND

The bond energies in part (c) are an average taken from a range of values in a range of similar compounds; [1 mark]

[Total: 2 marks]

4a

a)

i) The completed table to show the oxidation number of nitrogen in each compound:

Compound	NO	NO ₂
oxidation number of N	+2; [1 mark]	+4; [1 mark]

ii) The equations for the formation of NO₂:

- $\frac{1}{2}\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$;

OR



- $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$;

OR



[Total: 3 marks]

- Remember that to form an uncharged compound the oxidation numbers have to balance to zero
- For NO
 - Oxidation number of oxygen atom = -2
 - NO has no charge so the total oxidation number is 0
 - Oxidation number of N = $+2$
- For NO₂
 - Oxidation number of each oxygen atom = -2
 - For two oxygen atoms = $2 \times (-2) = -4$
 - NO₂ has no charge so the total oxidation number is 0

- Oxidation number of N = +4
- State symbols are not necessary to get the marks for the equations

4b

b) The bond enthalpy, in kJ mol^{-1} , of the N=O bond:

Bonds Broken / kJ mol^{-1}	Bonds Formed / kJ mol^{-1}
$2 \times \text{O}=\text{O} = 2 \times 495 = 990$ $1 \times \text{N}\equiv\text{N} = 1 \times 941 = 941$	$4 \times \text{N}=\text{O} = 4 \times ? = 4(\text{N}=\text{O})$
Total = +1931 [1 mark]	Total = -4(N=O)

- $\Delta H_f = +1931 - 4(\text{N}=\text{O}) = +497 \text{ kJ mol}^{-1}$; [1 mark]
- Bond enthalpy of the N=O bond = $\frac{1434}{4} = 358.5 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- The reaction equation is



- From this we can see that 1 $\text{N}\equiv\text{N}$ and two $\text{O}=\text{O}$ bonds are broken and 4 N=O bonds are formed
- The complete workings for the last step are here

$$\Delta H_f = +1931 - 4(\text{N}=\text{O}) = +497 \text{ kJ mol}^{-1}$$

$$-4(\text{N}=\text{O}) = +497 - 1931$$

$$4(\text{N}=\text{O}) = 1434$$

$$\text{Bond enthalpy of the N=O bond} = \frac{1434}{4} = 358.5 \text{ kJ mol}^{-1}$$

- Bond energies are usually expressed without a sign as they can be positive or

- Oxidation number of oxygen atom = -2
 - NO has no charge so the total oxidation number is 0
 - Oxidation number of N = +2
- For NO₂
 - Oxidation number of each oxygen atom = -2
 - For two oxygen atoms = 2 x (-2) = -4
 - NO₂ has no charge so the total oxidation number is 0
 - Oxidation number of N = +4
- State symbols are not necessary to get the marks for the equations

4b

b) The bond enthalpy, in kJ mol⁻¹, of the N=O bond:

Bonds Broken / kJ mol ⁻¹	Bonds Formed / kJ mol ⁻¹
2 x O=O = 2 x 495 = 990	4 x N=O = 4 x ? = 4(N=O)
1 x N≡N = 1 x 941 = 941	
Total = +1931 [1 mark]	Total = -4(N=O)

- $\Delta H_r = +1931 - 4(\text{N=O}) = +497 \text{ kJ mol}^{-1}$; [1 mark]
- Bond enthalpy of the N=O bond = $\frac{1434}{4} = 358.5 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- The reaction equation is



- From this we can see that 1 N≡N and two O=O bonds are broken and 4 N=O bonds are formed
- The complete workings for the last step are here

$$\Delta H_r = +1931 - 4(\text{N=O}) = +497 \text{ kJ mol}^{-1}$$

$$-4(\text{N=O}) = +497 - 1931$$

$$4(\text{N=O}) = 1434$$

$$\text{Bond enthalpy of the N=O bond} = \frac{1434}{4} = 358.5 \text{ kJ mol}^{-1}$$

- Bond energies are usually expressed without a sign as they can be positive or negative

4c

c) Suggest a reason for the difference in boiling points between carbon dioxide and nitrogen dioxide:

- The van der Waals/ intermolecular forces are (much) stronger in NO₂ / weaker in CO₂; [1 mark]

ONE from

- NO₂ has permanent dipole-dipole attractions (between the molecules) but CO₂ does not ; [1 mark]
- NO₂ is polar but CO₂ is not; [1 mark]
- In CO₂ the molecule is symmetrical and the dipoles cancel out / there is no net dipole; [1 mark]

[Total: 2 marks]

- This question shows how much shape influences intermolecular forces and boiling point
- The C=O bond is actually more polar than the N=O bond as there is a greater difference in electronegativity between the atoms, but the presence of a lone pair on N makes it bent linear, whereas CO₂ is linear
- The linear shape causes the dipoles to cancel out in CO₂, so there is no net dipole and hence it only has weaker dispersion(London) forces

4d

d) An equation for the enthalpy of formation of nitrogen monoxide, NO:

- $\frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}(\text{g})$; [1 mark]

[Total: 1 mark]

- Be careful- an easy trap to fall into!
- Enthalpy of formation is defined as the formation of *1 mole* of a compound from its elements in their standard states and weaker candidate would write it as:



- State symbols must be included to score the mark as they are part of the definition of enthalpy of formation

5a

a) The mass of propane required is:

- $q = mc\Delta T$ calculation
 $150 \times 4.18 \times 79 = 49533 \text{ J mol}^{-1}$ **OR** $49.533 \text{ kJ mol}^{-1}$; [1 mark]
- Moles of propane = $\frac{49.533}{2219.2} = 0.0223\dots$; [1 mark]
- Mass of propane = $0.023 \times 44.0 = 0.98 \text{ g}$; [1 mark]

[Total: 3 marks]

- This is another way of asking a $q = mc\Delta T$
- You have all the information to calculate q
 - Remember to use the mass of water that has been boiled
- The more challenging part is to then calculate the number of moles of propane used
 - q is calculated in J, so you must convert it to kJ before dividing it by $(-)\text{2219.2}$ which is the enthalpy of combustion of propane
 - Enthalpy of combustion is the energy released when 1 mole of the compound or element burns completely in a plentiful supply of oxygen
 - But we don't have one mole of propane, we have 0.0223.... moles
 - Therefore to find out the mass in grams, you must multiply by the M_r of propane, 44.0

5b

b) More propane is required than calculated in part (a) because:

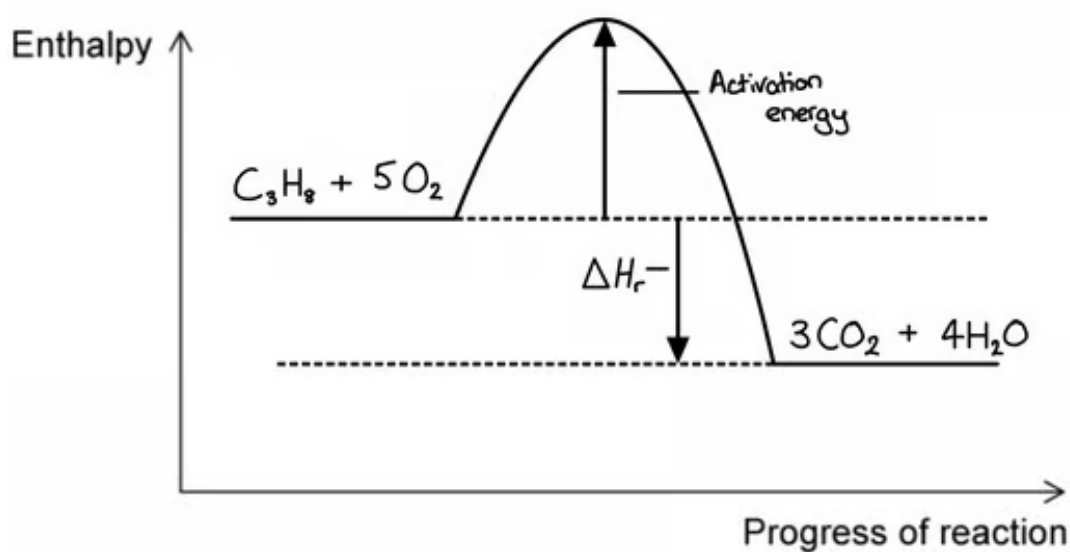
- Some of the heat / energy is lost to the surroundings; [1 mark]

[Total: 1 mark]

- This is a practical example so you should consider the experiments that you have done previously to help you answer this question
 - Heat loss to the surroundings is a common error in calorimetry experiments

5c

c) The completed diagram is:



- Correct location of and stoichiometry of reactants and products; [1 mark]
- Upward arrow labelled as activation energy; [1 mark]
- Downward arrow labelled as ΔH_r^- / enthalpy change / ΔH_c ; [1 mark]

[Total: 3 marks]

- The combustion of propane is an exothermic reaction, so the reactants must be higher in energy than the products
- Your reactants and products should be written as a balanced equation

